

A Minimal Symmetry-Defect Pressure for Reflection-Selected Hénon Packets

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Abstract

Primitive odd reflection points of the full Hénon horseshoe have two natural sampling limits: a marked reflection-boundary Bernoulli law and, after cyclic time averaging, the invariant maximal-entropy Bernoulli law. A direct rigorous separation of their coordinate-Mahler averages requires cylinder distortion estimates not presently available. We construct instead the minimal finite cross-axis calibration observable $\chi(\omega) = \mathbf{1}_{\{\omega_{-1}=\omega_1\}}$. Every finite observable supported wholly on one side of the axis has equal expectations under the two laws, but $\mathbb{E}_J\chi = 1$ and $\mathbb{E}_B\chi = 1/2$. The resulting extensive packet pressures are exactly $\frac{1}{2}\log 2 - t$ and $\frac{1}{2}\log 2 - t/2$. Coupling this calibration to the coordinate Mahler variable produces two pressure planes whose transverse derivatives differ by $1/2$, regardless of the unresolved Mahler-slope gap. Exact primitive-palindrome certificates through period 21 accompany the theorem. No prime trace or operator is claimed.

1 Sampling problem

The area-preserving map $H(x, y) = (6 - x^2 - y, x)$ has a real chain recurrent set conjugate to the full two-shift, with its reversor intertwined with sequence reversal. This rests on the Devaney–Nitecki horseshoe and Arai’s hyperbolic plateau [2, 1]; symbolic background is standard [3].

Let μ_B be fair two-sided Bernoulli measure. Let η_J be the reflection-boundary law obtained from iid fair bits $(\xi_k)_{k \geq 0}$ by

$$\omega_k = \omega_{-k} = \xi_k \quad (k \geq 0). \quad (1)$$

The preceding packet-equidistribution theorem proves that marked primitive reflection points converge to η_J , while cyclically averaging those points converges to μ_B .

For the coordinate observable $\varphi = \log^+ |x|$, these limits define constants κ_J and $\kappa_{\max}$. Finite data suggest they differ, but an all-period numerical inequality needs an explicit coding-distortion constant. We seek a rigorous local calibration that avoids that missing input.

2 One-sided blindness and the first separator

Call an observable positive-one-sided if it depends on finitely many coordinates in $\{0, 1, 2, \dots\}$; define negative-one-sided analogously.

Theorem 2.1 (One-sided blindness). *For every finite positive- or negative-one-sided observable f ,*

$$\int f d\eta_J = \int f d\mu_B. \quad (2)$$

Proof. Under both laws, every finite vector $(\omega_0, \dots, \omega_r)$ consists of independent fair bits. The negative side follows by reflection. $\square$

Define the centered radius-one observable

$$\chi(\omega) = \mathbf{1}_{\{\omega_{-1}=\omega_1\}}. \quad (3)$$

Proposition 2.2 (Minimal cross-axis separator). *One has*

$$\int \chi d\eta_J = 1, \quad \int \chi d\mu_B = \frac{1}{2}. \quad (4)$$

Thus a centered radius-one cross-axis window distinguishes the measures, whereas theorem 2.1 excludes every finite one-sided window.

Proof. Equation (1) makes the first equality deterministic. Under μ_B , the two tested bits are independent and fair. $\square$

The minimality statement is deliberately support-sensitive. It does not claim invariance under an arbitrary sliding-block recoding.

3 Finite primitive reflection packets

Let n be odd and let U_n be the $2^{(n+1)/2}$ palindromic words fixed by the marked reversal. Their exact-period subset has size

$$D_n = \sum_{d|n} \mu(n/d) 2^{(d+1)/2}. \quad (5)$$

At the marked center, (3) equals one for every word.

Now choose both a word in U_n and a cyclic center uniformly. At the reflection center the two tested sites are paired. At each of the other $n-1$ centers they are distinct free half-word bits. Hence the exact full mean is

$$c_n^{\text{full}} = \frac{1}{n} + \frac{n-1}{2n} = \frac{1}{2} + \frac{1}{2n}. \quad (6)$$

Removing lower periods changes expectations of a $[0, 1]$ -valued observable by at most

$$\tau(n) 2^{-n/3}. \quad (7)$$

Therefore the primitive orbit mean c_n tends to $1/2$.

4 Separated pressure planes

Define the pure symmetry packet partitions

$$Z_{n,J}(t) = D_n e^{-tn}, \quad Z_{n,\text{orb}}(t) = D_n e^{-tnc_n}. \quad (8)$$

Since $n^{-1} \log D_n \rightarrow \frac{1}{2} \log 2$, we obtain the following exact laws.

Theorem 4.1 (Symmetry-defect pressures). *For every fixed real t ,*

$$P_J(t) = \frac{1}{2} \log 2 - t, \quad P_{\text{orb}}(t) = \frac{1}{2} \log 2 - \frac{t}{2}. \quad (9)$$

Add the packet-Mahler parameter s from the preceding work. The resulting two planes are

$$P_J(s, t) = \frac{1}{2} \log 2 - s\kappa_J - t, \quad (10)$$

$$P_{\text{orb}}(s, t) = \frac{1}{2} \log 2 - s\kappa_{\text{max}} - \frac{t}{2}. \quad (11)$$

In particular,

$$\partial_t P_J - \partial_t P_{\text{orb}} = -\frac{1}{2}. \quad (12)$$

This is a certified transverse separation even though the value of $\kappa_J - \kappa_{\text{max}}$ remains open.

5 Executable certificate and boundary

Two independent programs enumerate primitive palindromes through periods 21 and 17, respectively. They verify (6), (7), the degree vector 2, 2, 6, 14, 28, 62, 126, 246, 510, 1022, 2030, and 21 hostile claim mutations. Normal and optimized tests agree. No floating-point input is used.

The theorem gives a symbolic calibration channel, not a solution of the coordinate-Mahler gap. Nor does it provide rational-prime labels, a von Mangoldt amplitude, a Fredholm determinant, a self-adjoint operator, or a Hilbert–Pólya correspondence. Its next structural question is whether the marked pressure is invariant under replacing a potential by a symbolic coboundary. Because η_J is not invariant, one should expect an exact boundary gauge anomaly.

References

- [1] Zin Arai. On hyperbolic plateaus of the h  non map. *Experimental Mathematics*, 16(2): 181–188, 2007. doi: 10.1080/10586458.2007.10128992.
- [2] Robert L. Devaney and Zbigniew Nitecki. Shift automorphisms in the h  non mapping. *Communications in Mathematical Physics*, 67:137–146, 1979. doi: 10.1007/BF01221362.
- [3] Douglas Lind and Brian Marcus. *An Introduction to Symbolic Dynamics and Coding*. Cambridge University Press, 1995. doi: 10.1017/CBO9780511626302.